Distributive concord with complex licensing

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Abstract

Distributive concord items are only grammatical when licensed in an appropriate plural or distributive environment. In this paper, we introduce a new phenomenon of *complex licensing*, in which the relevant semantic environment emerges from the interaction between two operators, neither of which is sufficient to license the item by itself. The relevant examples involve disjunction scoping below a distributive operator of a different ontological type. We show that these facts are predicted in a straightforward manner from semantic analyses of distributive concord, but pose significant challenges to syntactic analyses. Further comparisons are made between different theories of semantic licensing.

1 Introduction

Many morphological items in natural language have been shown to require *licensing* by a particular grammatical environment. On such example involves distributivity: in many cases, distributive numerals or pluractional verbs are only grammatical under a distributive or plural operator. A canonical paradigm is illustrated in (1) with the distributive numerals *ju-jun*, 'one-one', and *ox-ox*, 'three-three', of the Mayan language Kaqchikel. When the distributive numeral appears in the scope of a distributive or plural subject, as in (1a,b), the sentence is fine, but under a singular subject, as in (1c), the sentence becomes ungrammatical. One can thus identify the distributive or plural subject as the *licensor* of the distributive numeral in question. Because distributivity may be marked on multiple items in the sentence—e.g. *chikijujunal*, 'each', and *ju-jun* in (1a)—, such patterns have been sometimes called *distributive concord*.

(1) a. Chikijujunal ri tijoxela' xkiq'etej ju-jun tz'i'.
each the students hugged one-one dog
'Each of the students hugged a dog.'

b. Xeqatij ox-ox wäy
we-eat three-three tortilla
'We ate three tortillas each.'

c. * Xe'inchäp ox-ox wäy.
I-handle three-three tortilla
'I took three tortillas each.' (Kaqchikel, Henderson 2014)

One theoretical thread of work has pursued the hypothesis that these patterns should be explained in terms of *semantic licensing*: the grammaticality of the distributive concord item depends on the semantic properties of the environment in which it appears—roughly speaking, it must generate a plurality of entities (Henderson, 2014; Kuhn, 2017). An alternative hypothesis, however, explains these patterns in terms of *syntactic licensing* (Rushiti and Dobrovie-Sorin, 2023). On such a theory, potential licensors bear a particular morphological feature; the distributive concord item must enter into a syntactic relation with such a feature to be grammatical.

Here, I present a new kind of grammatical configuration on which distributive concord items are licensed. For these cases, which I will call *complex licensing*, the licensing environment is not generated by a single operator, but rather, by the interaction of two elements of the sentence, neither of which is sufficient to license the item by itself.

The specific configuration discussed here arises when a disjunction appears in the scope of a distributive quantifier. In such configurations, I show that a parallel paradigm emerges for two distinct phenomena in two unrelated languages: first, for participant-key pluractionals in French Sign Language (LSF); second, for the internal reading of *same* in English. In (2), neither TOUS-LES-JOURS, 'every day' nor disjunction (here, PU) is sufficient to license the pluractional marker '-alt' by itself, but together, they create an environment that does. In (3), neither *would* nor *or* is sufficient to license the internal reading of *same* by itself, but together, they create an environment that does.

(2) a. * TOUS-LES-JOURS JEAN VIENT-alt
every-day Jean come-alt
'Every day, Jean comes.'

b. * IX1 SENT, DEMAIN JEAN PU PIERRE PU MARIE VIENT-alt
I suspect tomorrow Jean or Pierre or Marie come-alt
'I suspect that tomorrow, Jean, Pierre, or Marie will come.'

c. TOUS-LES-JOURS JEAN PU PIERRE PU MARIE VIENT-alt
every-day Jean or Pierre or Marie come-alt
'Every day, Jean, Pierre, or Marie comes.

(3) a. * I would give Leon the same answer.

b. * She gave Leon or Marta the same answer.

c. I would give Leon or Marta the same answer. *(On internal reading)*

The existence of complex licensing provides an argument in favor of semantic theories of licensing. On a semantic analysis, what matters are the semantic properties of the environment in which the distributive concord item appears, regardless of how these properties are generated compositionally. In contrast, on a syntactic analysis, licensing is due to a morphological feature residing somewhere in the structure; but, in the examples above, it is impossible to identify any item as 'the licensor', which could bear the relevant feature, without important further stipulations. In the remainder of the paper, I lay out this argument, and provide explicit explicit compositional semantic analyses of distributive concord with complex licensing.

The unique properties of complex licensing provide a new source of data to adjudicate between different theories of distributive concord. First, as sketched above, the phenomena provides strong evidence in favor of a process of semantic licensing. In Section 4, an analysis is proposed on which it is in fact possible to identify one syntactic item as 'the licensor', potentially providing a lifeline to the syntactic analysis; nevertheless, any analysis must *additionally* retain a process of semantic licensing.

Second, the phenomenon provides an interesting point of comparison for different frameworks *within* theories of semantic licensing. In Sections 2-3, a compositional analysis is provided within event semantics, following work on licensing of pluractionals (Kuhn and Aristodemo, 2017) and providing for perhaps the first time a compositional implementation of Carlson's (1987) event-semantic analysis of *same*. In Section 4, in order to capture a larger body of data, an architectural revision of the analysis is proposed in terms of parasitic scope (Barker, 2007). Comparison of the two analyses provides insight into the relation between the two frameworks (what they can and can't capture) and into the formal properties that are shared between both analyses, including, notably, a central role for semantic scope.

2 Distributive concord

The simplest way to introduce a plural in natural language is with plural marking; in English, for example, the suffix *-s* transforms singular *zebra* to plural *zebras*, which includes in its denotation all pluralities generating by summing the atomic elements in *zebra*. But languages also have ways of generating pluralities indirectly through the compositional semantics. In (4a), for example, the noun *book* has singular marking, but one nevertheless infers the existence of a plurality of books—one per student. In (4b), the sentence similarly entails a plurality of events—one arrival per friend. These pluralities are generated because the noun or verb in question appear in the scope of a distributive operator, *each*.

(4) a. The students each read one book.
    b. My friends each arrived.

Cross-linguistic work has shown that a wide variety of non-English languages have morphological strategies to mark these derived pluralities. When this marking appears on noun phrases, the resulting items are called *distributive numerals* or dependent indefinites (Gil, 1982; Farkas, 1997). In (1a), for example, the noun phrase *ju-jun tz'i'*, 'one-one dog', appears in the singular, but the reduplication of the numeral indicates that a plurality of dogs is entailed when the sentence is evaluated as a whole. Similar patterns have been shown to hold for what is called *pluractional* marking on verbs, indicating a plurality of events (Newman, 2012; Cusic, 1981). In the LSF sentence in (5), for example, the pluractional marker '-alt' entails a plurality of arriving events, and is licensed by the plural subject AMI POSS-1 IX-pl, 'my friends'.

(5) AMI POSS-1 IX-pl ARRIVE-alt
    friend my they arrive-alt
    'My friends each arrived.' (LSF, Kuhn and Aristodemo 2017)

Distributive concord necessarily involves two pluralities: the plurality introduced by the distributive concord item—called the *share*—, and the plurality with respect to which it is distributed—called the *key*. In (5), for example, the share is the plurality of arriving events and the key is the plurality of friends: there is one arrival per friend. One fact that will be important for the development of the argument below is that distributive concord items may place ontological restrictions on what can serve as a potential key. This is illustrated with two pluractional markers in LSF, '-rep' (exact reduplication) and '-alt' (alternating reduplication of the two hands), shown in (6). While both forms of OUBLIE, 'forget', entail a plurality of forgetting events, they differ in what can serve as a distributive key. OUBLIE-rep entails that these events are distributed across time (it requires a temporal key); OUBLIE-alt entails that the events are distributed across participants (it requires a participant key). The pluractional marker '-alt' would thus not be grammatical in the frame in (6a), in which both subject and object are singular.

(6) a. UN PERSONNE OUBLIE-rep UN MOT
one person forget-rep one word
'One person forgot one word repeatedly.'

b. UN PERSONNE OUBLIE-alt PLUSIEURS MOT
one person forget-alt several word
'One person forgot several words.' (LSF, Kuhn and Aristodemo 2017)

A compositional puzzle discussed in the literature regards the fact that distributive concord items can be licensed by distributive operators. Distributive quantifiers like *each* entail that the property in their scope holds of each atomic element in their restrictor. In Kaqchikel, for example, the distributive quantifier *chikijujunal*, 'each', distributes to atomic individuals; as a consequence, the collective verb *xkimol*, 'gather', is infelicitous in its scope in (7a), since it is infelicitous to assert that an atomic individual gathered. In contrast, despite the fact that predicates with distributive numerals are similarly ungrammatical if the subject denotes an atomic individual, they turn out to be grammatical under distributive operators, as seen above in (1a), repeated as (7b).

(7) a. * Chikijujunal xkimol ki' pa k'ayb'al.
each gathered REFL in market
'Each of them gathered in the market.'

b. Chikijujunal ri tijoxela' xkiq'etej ju-jun tz'i'.
each the students hugged one-one dog
'Each of the students hugged a dog.' (Kaqchikel, Henderson 2014)

A parallel situation has been described for pluractionals. As described above, in LSF, the pluractional marker '-alt' is ungrammatical with only singular arguments; nevertheless, it can be licensed by a distributive operator like CHACUN, 'each', as in (8a). (As in the Kaqchikel example, collective predicates like 'gather' are ungrammatical under CHACUN.) The pluractional marker '-rep' shows similar pattern, exemplified in (8b). Typically, the temporal quantifier TOUS-LES-JOURS asserts that the predicate in its scope holds of each day. While the sentence (8b) has a reading in which each day involves an event of repeatedly giving books, it also has a reading—the preferred reading—in which a single book is given each day.

(8) a. GARÇON CHACUN OUBLIE-alt APPORTE APPAREIL-PHOTO
boy each forget-alt bring camera
'Each boy forgot to bring a camera.'

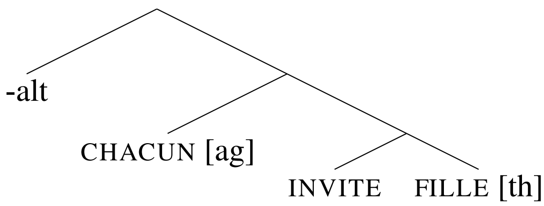
b. TOUS-LES-JOURS UN LIVRE JEAN DONNE-1-rep
every-day one book Jean give-me-rep
i. ‘Every day, Jean gave me one book.’
ii. ‘Every day, Jean gave me books repeatedly.’
(LSF, Kuhn and Aristodemo 2017)

In the semantic literature, a solution to this compositional puzzle has been proposed in terms of scope-taking (Henderson, 2014; Kuhn and Aristodemo, 2017; Kuhn, 2017). Within the scope of the distributive operator, a single event or individual is introduced, but outside of the scope of the distributive operator, these events or individuals are summed to create a plurality. On a semantic analysis, the distributive concord item takes scope above the distributive operator to access this plurality. Examples (9)–(11) illustrate such an analysis for the pluractional marker ‘-alt’ in LSF (Kuhn and Aristodemo, 2017). The definition of ‘-alt’ in (9) entails that sub-events have different participants—$\Theta(e)$ is defined as the tuple of the arguments of an event: $\langle \text{agent}(e), \text{theme}(e), \ldots \rangle$. Distributive operators like CHACUN are taken to sum events to create event pluralities (Champollion, 2016). When licensed by the distributive operator CHACUN, ‘-alt’ is interpreted at a higher position, thus accessing the plurality of events created by CHACUN, as shown in (11).

(9) $[\![\text{-alt}]\!] = \lambda V e[V(e) \wedge \exists e', e'' \preceq e[\Theta(e') \neq \Theta(e'')]]$

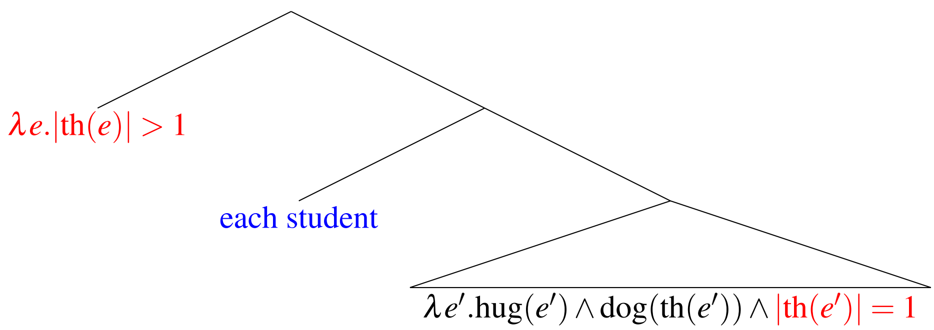
(10) CHACUN INVITE-alt FILLE
each invite-alt girl
‘Each of them invited a girl.’ (LSF, Kuhn and Aristodemo 2017)

(11)



Distributive numerals pose a further compositional challenge, as they appear to be interpreted both *high* and *low* in the logical form, thus necessitating a mechanism of *split scope*. The situation can be illustrated with the Kachikel distributive numeral *ju-jun*, ‘one-one’. Intuitively, *ju-jun* contributes two cardinality checks: below the distributive operator, it checks that there is one entity; above the distributive operator, it checks that there is more than one entity. This is intuitively illustrated in the pseudo-LF in (12), in which the distributive numeral contributes meaning (shown in red) both above and below ‘each boy’.

(12)

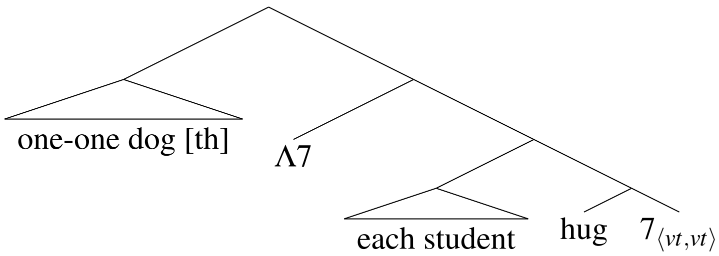


To provide a compositional semantics for such examples, an analysis must thus provide a mechanism of split scope, as well as a mechanism of indexing to ensure that the plurality of individuals evaluated above the distributive operator correspond to the atomic individuals introduced below the distributive operator. In existing semantic analyses of distributive numerals, this latter task is achieved by the use of a dynamic semantics of plurals (Henderson, 2014; Kuhn, 2017). The individuals under the distributive operator are introduced at some index i in separate assignment functions; the same index is then preserved when the assignment functions are summed by the distributive operator.

An identical effect can be achieved in event semantics by indexing discourse referents via their thematic role.[1] In event semantics, thematic roles are standardly assumed to show *cumulativity*: $\theta(e+e') = \theta(e) + \theta(e')$. As a consequence, a distributive numeral can tether the two cardinality checks by associating them with the same thematic role. An event-semantics definition for a distributive numeral is thus provided in (13). Observe that the cardinality check that $\theta(e') = 1$ appears within the scope of κ, while the cardinality check that $\theta(e') > 1$ appears outside of it. Compositionally, we implement split scope using quantifier raising with higher-order traces (Charlow, 2018; Kuhn, 2022). In the logical form in (14), an event predicate modifier (type $\langle vt, vt\rangle$) is raised over an event predicate (type vt); the type of 'one-one dog [th]' is thus $\langle\langle vt, vt\rangle, vt\rangle$.

(13) $[\![\text{one-one dog}]\!] = \lambda\theta\kappa e.\kappa(\lambda Ve'.V(e') \wedge \text{dog}'(\theta(e')) \wedge |\theta(e')| = 1)(e) \wedge |\theta(e)| > 1$

(14)



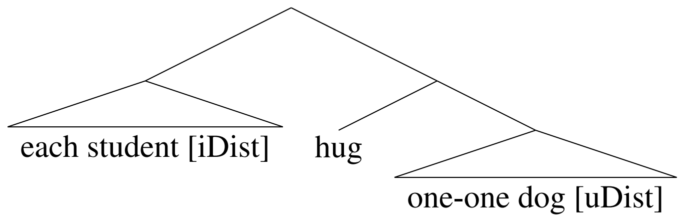
The definitions and logical forms above thus spell out an analysis of distributive concord in terms of semantic licensing, in which distributive concord items contribute a semantically interpreted plurality condition that may be interpreted at a higher level.

As described in the introduction, however, an alternative hypothesis explains distributive concord in terms of syntactic licensing. Notably, Rushiti and Dobrovie-Sorin (2023), observing parallels between distributive concord and negative concord, propose that the former be analyzed using mechanisms of feature checking that are assumed on a common analysis of the latter (Zeijlstra, 2004).[2] On such a theory, distributive concord items carry an uninterpreted feature, [uDist], and must be c-commanded by a higher operator carrying an interpreted [iDist] feature, as shown in (15). On such a theory, the meaning of a distributive numeral like *ju-jun*, 'one-one', is much simpler: it is identical to the meaning of the plain numeral *jun*, 'one'. The distributive interpretation and plurality inference arise entirely from the operator bearing the [iDist] feature.

[1] See Henderson (2014, p. 40) for a related proposal on which pluractional markers take thematic role operators as arguments, modifying the way they associate events and their participants.

[2] If convinced by the arguments that distributive concord and negative concord should be analyzed on a par, another possibility is, of course, to extend a theory of semantic licensing to negative concord, as pursued in Kuhn (2022).

(15)



3 Complex licensing of distributive concord

On a theory of semantic licensing, there is notably no reference to any particular item as *the* licensor; rather, the environment in which a distributive concord item appears must simply have certain semantic properties. This environment can thus in principle result from the interaction of two operators, neither of which can license the item alone. Such an interaction is not predicted on a syntactic theory of licensing, which requires the relevant syntactic feature to reside on a particular node in the structure.

In what follows, I show that such cases of *complex licensing* are indeed attested, in the interaction of disjunction with distributive operators.

3.1 Methods

LSF judgments reported below come from a single Deaf native signer using the playback method (Schlenker, 2011). In a first step, through conversation in LSF between the author and the signer, the signer was invited to produce a particular linguistic structure (e.g. “How would you use DONNE-alt in a sentence?”, “What about if you replace the plural subject with JEAN OU PIERRE?”). These productions were video recorded, and the signer was then asked to give their personal judgment on the naturalness of the sentence on a scale from 1 to 7 (7 best). The signer was welcome to discuss the sentences that they produced, either by giving metalinguistic comments about sentences (e.g. “Why would you use VIENT-alt if there are only two people?”), or by proposing alternative formulations that would be more natural, which were also recorded and judged. Video recordings were then shown to the same signer on three separate occasions to replicate naturalness judgments. Below, we report the numerical average of these three judgments.

English judgments are the author’s own judgments, discussed with several other native English speakers, and are reported with standard markers of grammaticality (e.g. *).

3.2 Pluractionals in LSF

Recall from example (6) above that distributive concord items may place ontological restrictions on what can serve as a possible licensor. As a consequence, the sentence in (16) is ungrammatical because the pluractional marker ‘-alt’ requires a participant key: the temporal quantifier TOUS-LES-JOURS, ‘every day’ is thus not sufficient to license it.

(16) [2] TOUS-LES-JOURS JEAN VIENT-alt
every-day Jean come-alt
‘Every day, Jean comes.’ MVI_2344/MVI_4244

We observe further that disjunction alone cannot license '-alt'. In LSF, there are at least two markers of disjunction: the sign OU involves two O-handshapes moving in alternating circles; the sign glossed as PU is a palm up gesture. In (17), the sign PU is used because repetition of OU was judged as 'heavy' in examples with three disjuncts. With this irrelevant source of unnaturalness controlled for, the example in (17), with '-alt', still receives a low judgment. Intuitively, the reason is because disjunction—essentially existential quantification over the listed domain—introduces only a single event participant. While LSF OU/PU, like English *or*, allows an inclusive reading, the sentence in (17) implicates that the event in question involves a single individual coming. The plurality condition of '-alt' is therefore not met.

(17) $^{1.7}$ IX1 SENT, DEMAIN JEAN PU PIERRE PU MARIE VIENT-alt
I suspect tomorrow Jean or Pierre or Marie come-alt
'I suspect that tomorrow, Jean, Pierre, or Marie will come.' MVI_2348

Notably, though, it turns out that the combination of these two operators *does* license '-alt', as shown in (18). On a theory of semantic licensing, the reason for this interaction is quite straightforward. When disjunction takes scope below the universal quantifier, the sentence generates the inference that sometimes Jean comes, sometimes Pierre, and sometime Marie. When the operator TOUS-LES-JOURS sums these events together, the resulting event plurality (distributed over time) consequently involves a plurality of individuals: the sum of Jean, Pierre, and Marie. This is exactly the semantic property necessary to license '-alt'.

(18) $^{4.7}$ TOUS-LES-JOURS JEAN PU PIERRE PU MARIE VIENT-alt
every-day Jean or Pierre or Marie come-alt
'Every day, Jean, Pierre, or Marie comes. MVI_2349

The same paradigm is replicated below with the verbs OUBLIE, 'forget', and DONNE, 'give'. In (20a) and (21a), the verb OUBLIE take a disjunction as subject; in (20b) and (21b), it takes a disjunction as direct object. As above, disjunction alone is unable to license '-alt' in (20), but disjunction combined with the temporal quantifier TOUS-LES-JOURS is able to license '-alt' in (21). Differently from the example with VIENT in (16), the example with a temporal quantifier (but no disjunction) in (19) already has a relatively high average rating—although still lower than the examples with the both the temporal quantifier and disjunction. When asked directly why '-alt' can be used in such examples, the signer explained that Jean might be forgetting homework assignments from a variety of different classes: math, French, etc. In this case, we can view DEVOIR, 'homework', as an indefinite; variation of this indefinite under the temporal quantifier can thus introduce a plurality of homework assignments and satisfy the semantic constraint of '-alt'. This is not possible for with VIENT in (16), which only has a single argument that is filled by a proper name.

(19) a. $^{4.3}$ TOUS-LES-JOURS JEAN OUBLIE-alt DEVOIR
every-day Jean forget-alt homework
'Every day, Jean forgets his homework.' MVI_2341/MVI_4251

(20) a. $^{2.3}$ IX1 SENT, DEMAIN JEAN OU PIERRE OUBLIE-alt DEVOIR
I suspect tomorrow Jean or Pierre forget-alt homework
'I suspect tomorrow Jean or Pierre will forget their homework.' MVI_2211

b. [1] IX1 SENT, DEMAIN JEAN OUBLIE-alt REGLE PU CRAYON PU GOMME
I suspect tomorrow Jean forget-alt ruler or pencil or eraser
'I suspect tomorrow Jean will forget his ruler, pencil, or eraser.' MVI_2356

(21) a. [6] TOUS-LES-JOURS JEAN OU PIERRE OUBLIE-alt DEVOIR
every-day Jean or Pierre forget-alt homework
'Every day, Jean or Pierre forgets their homework.' MVI_2212

b. [6] TOUS-LES-JOURS JEAN OUBLIE-alt REGLE PU CRAYON PUT GOMME
every-day Jean forget-alt ruler or pencil or eraser
'Every day, Jean forgets his ruler, pencil, or eraser.' MVI_2357

For DONNE, 'give', examples (22)-(24) show a similar pattern of results to OUBLIE. When asked about the relatively high judgment of (22), the signer indicated that '-alt' can be used when a person gives *a lot* of money, an inference that is generated by the use of the temporal quantifier TOUS-LES-JOURS. As above, this appears to arise due to the interaction of the temporal quantifier with an indefinite argument (here, the mass noun ARGENT, 'money').

(22) [4.3] TOUS-LES-JOURS JEAN DONNE-1-alt ARGENT
every-day Jean give-me-alt money
'Every day Jean gives me money.' MVI_2343/MVI_4234

(23) [2.3] IX1 SENT, DEMAIN JEAN OU PIERRE DONNE-1-alt ARGENT
I suspect tomorrow Jean or Pierre give-me-alt money
'I suspect that tomorrow, Jean or Pierre will give me money.' MVI_2222

(24) [6] TOUS-LES-JOURS JEAN OU PIERRE DONNE-1-alt ARGENT
every-day Jean or Pierre give-me-alt money
'Every day Jean or Pierre gives me money.' MVI_2223

Below, we offer a compositional analysis to capture the intuitions above, using the example in (18) for illustration. In (25), we provide a simple definition of the temporal quantifier TOUS-LES-JOURS that pluralizes an event predicate and ensures that parts of the plural event occur on each day. Standard event semantic denotations are provided for the predicate and for the disjunctive subject in (26) and (27). (28) repeats the definition for '-alt' provided above in (9), and (29) provides an LF.

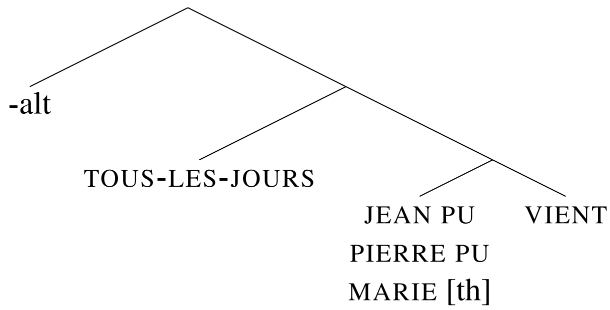
(25) $[\![\text{TOUS-LES-JOURS}]\!] = \lambda V \lambda e\,.\,{}^{*}V(e) \wedge \forall d[\text{day}(d) \rightarrow \exists e' \preceq e[\tau(e') \preceq d]]$
'e is a plurality of V-ing events whose runtime $\tau(e)$ overlaps with each day.'

(26) $[\![\text{VIENT}]\!] = \lambda e\,.\,\text{come}'(e)$
'e is a coming event.'

(27) $[\![\text{JEAN PU PIERRE PU MARIE [th]}]\!]$
$= \lambda V e\,.\,V(e) \wedge (\text{th}(e) = \text{jean}' \vee \text{th}(e) = \text{pierre}' \vee \text{th}(e) = \text{marie}')$
'e is a V-ing event whose theme is Jean, Pierre, or Marie.'

(28) $[\![\text{-alt}]\!] = \lambda V e[V(e) \wedge \exists e', e'' \preceq e[\Theta(e') \neq \Theta(e'')]]$
'e is a V-ing event whose subparts have different thematic arguments.'

(29)



Critically, when TOUS-LES-JOURS takes scope over the disjunctive subject, it creates an event plurality with a plural theme: some subparts have Jean as their theme, some have Pierre as their theme, and some have Marie as their theme. This is exactly the condition needed to license '-alt'.

3.3 *Same* in English

Turning now to English *same*, we first observe that *same*—despite ultimately entailing the uniqueness of a given individual—has pluractional properties. This can be seen in a variety of different ways. First, several authors have observed that *same* entails a plurality of events. Most prominently, Carlson (1987) observes that some readings of *same* are dependent on a distributive interpretation of the sentence, illustrated here by the contrast in (30). Sentence (30a), with a distributive reading, is totally fine, while (30b) cannot be used with a collective reading, involving a single event in which the two people co-authored a book. A similar observation holds for sentence (31) (and related examples from Carlson), which cannot be used to describe a single event in which John killed a man by hitting him. Finally, Hardt et al. (2012) observe that the sentence in (32) can only be used to described a situation in which a particular book was read (at least) twice, in contrast to an otherwise equivalent sentence in which *the same book* is replaced by the pronoun *it*, which is compatible with a single reading event.

(30) a. Irene and Angelika read the same book.

b. * Irene and Angelika wrote the same book.

(31) John hit and killed the same man. (Barker, 2007)

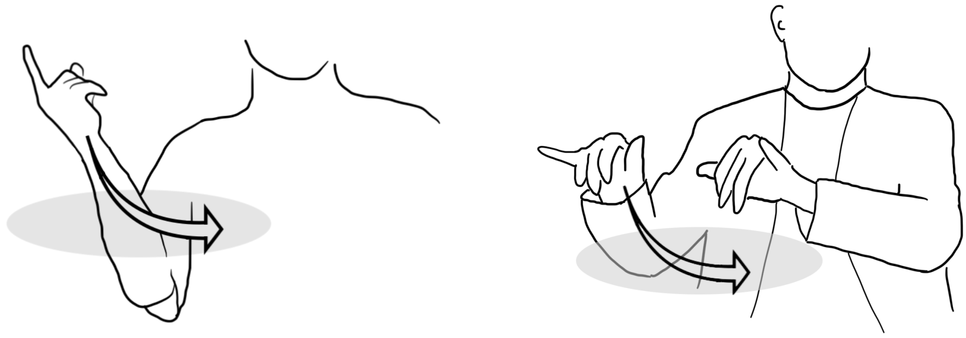
(32) John read a book last week, and he read the same book in a single go. (Hardt et al., 2012)

Two kinds of uses of *same* have been described in the literature, each exemplified in the sentences above. On its 'external' reading, seen in (32), *same* is anaphoric to another event that has been introduced in discourse and compares an individual in the asserted event to the corresponding individual in the other event. On the 'internal' reading of *same*, seen in (30) and (31), this comparison is done between the entities in a single sentence. On its internal reading, (30a) thus compares the books read by Irene and Angelika *to each other*. We note that (30a) also has an external reading—for example, if it were to follow the sentence *Barbara read a book*—in which the books read by Irene and Angelika are compared to the one read by Barbara.

The semantic environments in which the internal reading of *same* is available provides a second argument in favor of shared compositional properties with pluractionals. Recall from Section 2 that distributive numerals can be licensed by plurals, but also—unlike collective predicates—can be licensed by distributive operators like *each*. Kuhn (2017, 2019) observes that exactly the same pattern of licensing holds for the internal reading of *same*. Both (33a) and (33b) allow an internal reading, in which the answers of the different students are compared to each other. Sentence (33c) is not ungrammatical *per se*, but, uttered out of the blue, elicits a response "As who?", indicating that the reading accessed is the external reading. Here and below, we use the * to indicate the inability of the sentence to express the internal reading.

(33) a. The students gave the same answer.
 b. Each student gave the same answer.
 c. * Edith gave the same answer.

Third, Kuhn (2017) shows that in American Sign Language, the internal reading of *same* has same (iconic) morphology as distributive numerals. In both cases, the signs show plural morphology, moving over an area of space associated with the item's plural or distributive licensor.



(34) BOY IX-arc-a READ {ONE/SAME}-arc-a BOOK.

In light of these similarities, Kuhn (2017) thus analyzes *same* using the same tools as distributive numerals: low in the structure, *same* introduces a discourse referent at an index x; high in the structure, it presupposes a plurality of events, and asserts identity of x across these events.

Finally, we observe that, like distributive numerals and pluractionals, the internal reading of *same* places ontological restrictions on what pluralities can function as a licensor. Examples (35) and (36) show that *same* can be licensed by a plurality of individuals, and by a plurality of times. In contrast, (37) shows that it cannot be licensed by a plurality of worlds. (Note that the intended reading of (37a) can nevertheless be summarized quite easily as 'There is a particular poem that I have to read.')

(35) The students read the same poem.

(36) I always read the same poem.

(37) a. * I have to read the same poem.
 b. * I would read the same poem.

With this background, we now turn to a case of complex licensing that parallels that of the LSF case above. Like pluractionals in LSF, the disjunction *or* is not sufficient to license *same* by itself, as seen in (38). Further, as we have just observed, *would* introduces a plurality of worlds, but this is not sufficient to license *same*, as shown in (39). But, when *would* scopes over *or*, the internal reading of *same* becomes available, as shown in (40).

(38) * Alexis gave Leon or Marta the same answer.

(39) * I would give Leon the same answer.

(40) I would give Leon or Marta the same answer.

Notably, licensing depends not only on the presence of both components, but also their scopal configuration. Specifically, (40) becomes ungrammatical if the disjunction scopes above the distributive item. An example is provided in (41). Here, sluicing ensures that the disjunction is interpreted with highest scope. The sentence consequently becomes ungrammatical on an internal reading of *same*. (It remains grammatical on an external reading—for example, if it has already been mentioned that an answer was given to Nicole.)

(41) * Alexis would give Leon or Marta the same answer, but I don't know which one of them.

Intuitively, the reason for this pattern is very similar to the LSF pattern above. The modal *would* is a universal quantifier, witnessed by the fact that it cannot be used to conjoin mutually incompatible propositions, in contrast to existential modals like *could*, as shown in (42). When this universal takes scope over *or*, it induces variation, thus creating a plurality of individuals at a higher level of evaluation. Sentence (40) thus entails the possibility that an answer is given to Leon and and that an answer is given to Marta.

(42) a. * I would stay and I would leave.
b. I could stay and I could leave.

The data with *same*, however, nevertheless present some subtleties that go beyond the LSF example. On the one hand, we make the correct prediction that complex licensing of *same* should not be possible with existential modals, as the interaction with disjunction does not generate a plurality of individual discourse referents. This prediction appears to be borne out, as seen in (43).

(43) a. ? I could give Leon or Marta the same answer.
b. * I can give Leon or Marta the same document.

On the other hand, the logic we have just sketched suggest that a pattern of complex licensing should arise for any universal modal that takes scope over disjunction. This, however, is not the case, as shown in (44): the sentence cannot be used to express that there is a particular document that I have to give to one of the two mentioned people.

(44) * I have to give Leon or Marta the same document.

In order to understand the source of the difference between *would* and *have to*, a useful heuristic is to rewrite the the sentence in question with a semantically equivalent paraphrase, as is presented in (45) for *would*. Notably, when the same process is attempted for *have to*, the resulting paraphrase is a presupposition failure, as the first half of the sentence carries the inference that there isn't a document that I have to give to Leon, nor a document that I have to give to Marta.

(45) a. I would give Leon or Marta the same answer. $\rightarrow$
b. 'I would give Leon or Marta an answer, and the answer I would give to Leon and the answer I would give to Marta are the same.'

(46) * a. I have to give Leon or Marta the same document $\rightarrow$
b. 'I have to give Leon or Marta a document, and the document I have to give to Leon and the document I have to give to Marta are the same.'

More precisely, we propose that the difference between (40) and (44) arises from an additional presupposition of *same* and from somewhat unusual properties of the lexical semantics of *would* that distinguish it from other universal modals. In particular, we observe that disjunction under *have to* gives rise to the inference that the disjunct alternatives are false, as shown in (47). In contrast, this inference does not arise under *would* (which in fact even generates the opposite inference, that they are true), as seen in (48).

(47) I have to give the document to Leon or Marta.
$\rightsquigarrow$ I don't have to give the document to Leon and I don't have to give it to Marta.

(48) I would help Leon or Marta.
$\rightsquigarrow$ I would help Leon and I would help Marta.

The inference seen in (47) is standardly analyzed as a scalar implicature, arising via competition with the logically stronger alternatives with a single disjunct (cf. Chierchia et al., 2012). Empirically, it appears that such strengthening does not take place for disjunction under *would*. While providing a full explanation for this observation falls outside the scope of this paper, we can offer the sketch of an analysis. In particular, we observe that *would* appears to require a covert or overt domain restriction that is not necessary for *have to*. In (48), for example, we implicitly understand the modal to be quantifying over some subset of contextually salient worlds—perhaps those in which help is needed or requested, as paraphrased in (49). Critically, if the implicit domain restriction itself contains disjunction or is anaphoric to a disjunctive discourse referent, as in (49), then disjunction appears in *both the restrictor and the scope* of the modal quantifier. Because the restrictor is downward entailing and the scope is upward entailing, replacing disjunction with a more specific alternative does not in fact generate a stronger proposition, so no scalar implicature is generated.

(49) I would help Leon or Marta *if they (=Leon or Marta) asked me to*.

The second piece of the explanation lies in the presuppositions of *same*. Specifically, *same* presupposes that the given predicate holds of each atom in the licensing plurality. Sentence (50) presupposes that Leon read a book and that Marta read a book, then asserts that these books are identical.

(50) Leon and Marta read the same book.
Presupposes: Leon read a book and Marta read a book.

As was informally observed in the paraphrases in (45) and (46), this presupposition generates a contrast for the target sentences. Sentence (40) carries the presupposition that there is an answer that I would give to Leon and that there is an answer that I would give to Marta, and then asserts that these two answers are identical. Similarly, sentence (44) carries the presupposition that there is a document that I have to give to Leon and that there is a document that I have to give to Marta. However, this presupposition contradicts the scalar implicature observed in (47), and *same* is consequently infelicitous.

Below, we offer a compositional analysis of the general pattern. First, (51) provides a simple definition of *would* that mirrors the definition of TOUS-LES-JOURS in (25), ensuring that parts of the plural event occur in each world in the relevant modal domain. A simple definition of a disjunctive argument is provided in (52). Notably, when *would* takes scope over *or*, as in (53), the resulting meaning is a set of plural events. By cumulativity of thematic roles, these plural events include events whose theme is $\text{leon}' + \text{marta}'$, a plural individual. As we have seen, this is what is needed to license *same*.

(51) $[\![\text{would}]\!] = \lambda V \lambda e\,.{}^{*}V(e) \wedge \forall w[\textsc{dom}(w) \rightarrow \exists e' \preceq e[e' \preceq w]]$
'e is a plurality of V-ing events that has subparts in each world in the relevant modal domain DOM.'

(52) $[\![\text{Leon or Marta [th]}]\!] = \lambda Ve.V(e) \wedge (\text{th}(e) = \text{leon}' \vee \text{th}(e) = \text{marta}')$

(53) I would help Leon or Marta.

A definition of *same* is provided in (54). For expositional purposes, the definition does not encode the presupposition discussed above—that the predicate holds of each atom of the licensing plurality. Compositionally capturing this presupposition is in fact surprisingly complex, so we set it aside for present purposes, and return to it in Section 4.

We assume that *same* is anaphoric to a particular distributive key; in the definition below, we let π_i return a particular property of an event, which may be a thematic role or the runtime of the event. Anaphoricity allows *same* to compare events that differ with respect to this property; a compositional alternative with a similar effect in many cases is to allow *same* to take parasitic scope (Barker, 2007). Below, the presupposition in (54a) is one of event plurality, with the additional condition that these events vary with respect to the anaphorically-specified licensing plurality. The presupposition in (54b) provides the compositional spine of the definition, incorporating the meaning of the verb and noun in the scope of any licensing operators. Finally, (54c) asserts that the plurality of entities in at the relevant thematic role θ is the same across all atoms in the licensing plurality. (We use the notation $\sigma e[\phi]$ to indicate the sum of events e that have property ϕ.)

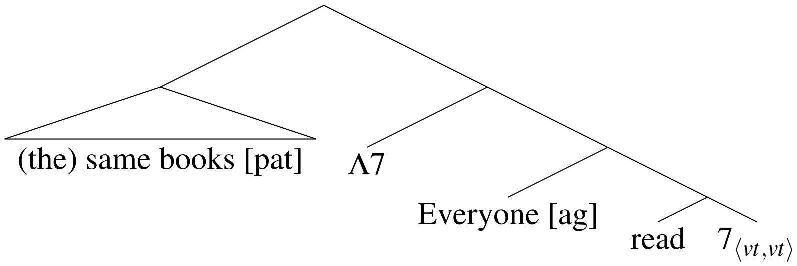
(54) $[\![\text{same}_i]\!] = \lambda N \theta \kappa e : \exists e', e'' \preceq e[\pi_i(e') \neq \pi_i(e'')] \wedge$ (a)
$\kappa(\lambda Ve'.V(e') \wedge N(\theta(e')))(e)\,.$ (b)
$|\{\theta(\sigma e'[e' \preceq e \wedge \pi_i(e') = x]) : x \prec_{\text{atom}} \pi_i(e)\}| = 1$ (c)

The definition is illustrated with the simple example in (55), assumed to have the logical form in (56). We observe that the definition in (54) and the logical form in (56) mirror the definition and logical form proposed for distributive numerals in (13) and (14).

For example (55), (54b) presupposes that *e* is an event in which everyone read books. We let π_i return the agent of an event; thus, (54a) presupposes that the agent of *e* is a plurality: there are multiple book readers. Finally, (54d) asserts that the plurality of books read is constant across each reader.

(55) Everyone read the same books.

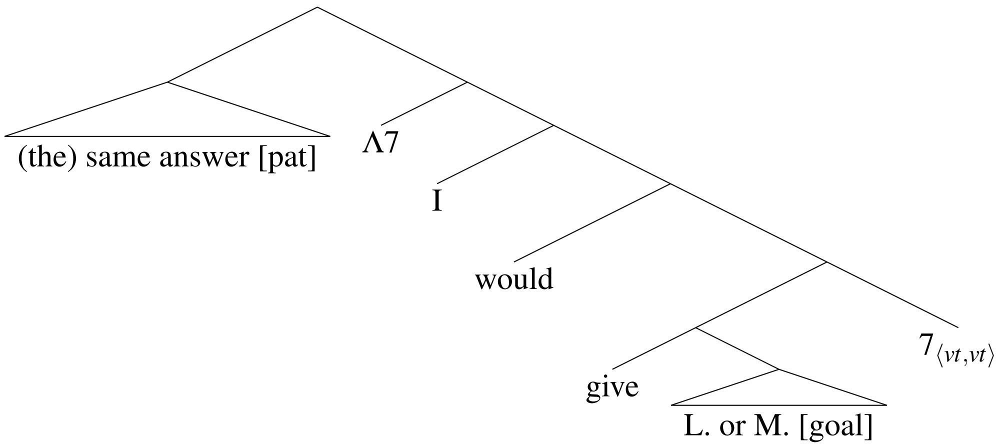
(56)



Turning to (57), assumed to have the logical form in (58), we apply the same logic. As above, (54b) presupposes that *e* is an event witnessing the fact that I would give Leon or Marta an answer. We let π_i return the goal of an event; (54a) thus presupposes that the goal of *e* is a plurality: multiple people receive answers. As we saw for example (53), when *would* takes scope over disjunction, the resulting predicate includes events that satisfy this predicate. Finally, (54d) asserts that the answers given are constant across each recipient.

(57) I would give Leon or Marta the same answer.

(58)



In Section 3.2, we derived a pattern of complex licensing for LSF '-alt' via the interaction of disjunction with a distributive temporal operator. On the present analysis of English *same*, a similar environment is generated by the interaction of disjunction and a universal modal; further constraints on the distribution of *same* arise from the way the presuppositions of *same* interact with scalar implicatures. Both the LSF and the English patterns provide evidence in favor of semantic licensing. On a syntactic analysis, it is impossible to identify any item as 'the licensor', which could bear the relevant syntactic feature. In contrast, on a semantic analysis, all we need is a plurality of individuals, regardless of how it is introduced.

3.4 Event semantics vs. dynamic semantics

As was discussed in Section 2, the present analysis relies on split scope, and requires a mechanism of indexing to ensure that the plurality of individuals above the distributive operator correspond to sum of the atomic individuals introduced below it. In prior literature, this semantic tethering has been done within the framework of dynamic semantics, in which discourse referents are attributed unique indices (Henderson, 2014; Kuhn, 2017). In the present paper, we have adopted a novel alternative, using event semantics, in which the semantic tethering is achieved using thematic role functions, assuming of cumulativity of thematic roles.

The reason for this choice is in part for convenience, due to the fact that both phenomena have pluractional properties, and the LSF example in particular is an inflection of a verb. The meanings are thus easily stated within an event semantics. But the choice is also in part a proof of concept. Notably, there is nothing fundamentally *dynamic* occurring in the phenomena in question (e.g. there are no cross-sentential dependencies). The present analysis thus shows that the same compositional strategy can be translated into a static event semantics, which has been used elsewhere in the literature on distributivity, including notably Lasersohn (1995) and Cable (2014), the latter of which explicitly leaves distributive concord as an open question.

This being said, in the literature on dynamic semantics, a similar compositional strategy has been used to account for a variety of other phenomena, including embedded definites (Bumford, 2017) and negative concord (Kuhn, 2022). It is unclear whether a parallel analysis can be constructed in event semantics to extend to these domains as well. Notably, in Kuhn (2022)'s analysis of negative concord, a negative concord item introduces a discourse referent low in the structure, then checks at a higher level that the extension of the relevant discourse referent is empty—a situation only arising when the item takes split scope around an anti-veridical operator like negation. Recent work on negation in event semantics has proposed the existence of negative events in order analyze sentences like (59) (Bernard and Champollion, 2024). However, this work has not yet been extended to describe what kind of object the theme of the negative event described in (59) is (i.e. what was eaten), and how it relates to the themes of the events in the event predicate that is negated. More work would thus need to be done on negation in event semantics to have a viable translation of the analysis in Kuhn (2022).

(59) I saw Mary not eat anything.

4 A revised analysis in terms of parasitic scope

In Section 3, we provided a relatively compact analysis of complex licensing for phenomena in two languages, pluractional verbs in LSF and the internal reading of *same* in English. The analysis was spelled out in event semantics, in which distributive operators sum together events witnessing a particular proposition, and the use of thematic roles allows one to relate the participants of the resulting sums to the participants of the component events. However, Section 3.3 notably left open one issue unresolved, regarding the inability of the combination of disjunction and *have to* to license the internal reading of *same*. A heuristic explanation was offered by means of paraphrase, but no compositional implementation was offered for this analysis sketch.

In the present section, we return to this open question, providing an explicit analysis of the relevant presuppositions of *same* and their interaction with exhaustivity. As we will see, however, the statement of these presuppositions requires a compositional machinery that is different in nature from the event-semantic analysis offered above. Specifically, the proposal will require *same* to take parasitic scope, as in Barker (2007) and unlike the event-semantic analysis provided above. The resulting semantic architecture will look fundamentally different than that of Section 3. Nevertheless, key elements remain constant: in particular, both turn on the semantic interaction of multiple operators—disjunction, a distributive operator, and, now, an exhaustivity operator.

Below, after providing further data to motivate the need for an additional presupposition of *same*, we demonstrate that an event-semantic analysis faces compositional challenges, but—perhaps surprisingly—so does an out-of-the-box analysis in terms of parasitic scope. We provide a new analysis in which parasitic scope is enriched with higher-order traces to allow semantic reconstruction. We show that this analysis provides room to account for further data, including the licensing of the internal reading of *same* in downward entailing contexts, and the variable availability of complex licensing with indefinites beyond disjunction.

4.1 The presuppositions of *same*

In Section 3.3, we proposed that *same* carries a presupposition that a particular property holds of each element of the licensing plurality. For our target sentence with *would*, repeated in (60), we proposed that the meaning is essentially equivalent to the paraphrase in (61). Notably, we take seriously the definite articles that appear in the paraphrase, such that (61)—and by assumption, (60)—carries the existential presupposition in (62).

(60) I would give Leon or Marta the same answer.

(61) The answer that I'd give to Leon and the answer I'd give to Marta are the same.

(62) There is an answer that I would give to Leon, and one that I would give to Marta.

As we have seen, this is at odds with an implicature generated when disjunction appear under *have to*. Specifically, the implicature in (63) contradicts the presupposition in (64).

(63) I have to send a document to Leon or Marta.
$\rightsquigarrow$ I don't have to send a document to Leon. (*Idem* for Marta.)

(64) There is a document that I have to send to Leon. (*Idem* for Marta.)

As it turns out, the inability for *have to* to participate in complex licensing is not specific to *have to*, nor is it specific to modal verbs. In particular, (65) provides an example in which the universal modal has been replaced by a universal temporal quantifier, thus creating an example parallel to the LSF case discussed in Section 3.2. Differently from previous examples, the relevant observation for (65) does not involve the availability of an internal reading *tout court*, since, as seen in (36), temporal quantifiers are themselves able to license an internal reading of *same*. Nevertheless, one can still investigate whether (65) allows complex licensing, since licensing by the temporal quantifier generates a different meaning—paraphrased in (65a)—than the licensing by the plurality of individuals that is constructed by the interaction of the temporal quantifier with the disjunction—paraphrased in (65b). As seen below, the meaning that would arise from complex licensing is not available to the sentence.

(65) Every day last week I gave Leon or Marta the same files.
   a. ✓ The same files were given every day, sometimes to Leon and sometimes to Marta.
   b. ✗ Potentially different files were given every day, but by the end of the week, Leon and Marta had been given the same files.

The unavailability of this reading is in fact predicted by the explanation offered above for *have to*. Once again, the implicature arising from the sentence in (66) contradicts the presupposition of *same* that appears in (67). In other words, the reading in (65a) is out for exactly the same reason that *have to* does not participate in complex licensing of *same*.

(66) I gave Leon or Marta files every day last week.
   $\rightsquigarrow$ I did not give Leon files every day last week. (*Idem* for Marta.)

(67) There are files that I gave Leon every day last week. (*Idem* for Marta.)

Replicating this pattern with a temporal quantifier illustrates several points. First, if complex licensing of *same* were only not available for *have to*, one could entertain the possibility that *have to* is not in fact a distributive operator, so simply does not generate the necessary plurality. The example in (65) shows that this is not a sufficient explanation, as the same pattern appears for *every day*, which certainly is a distributive operator. Second, these examples highlight how unusual the example with *would* is, where specific aspects of the lexical semantics permit this three-way interaction between the quantifier, the disjunction, and the mechanism that generates implicatures.

Finally, we have of course seen that temporal quantifiers do participate in the complex licensing of the distributive concord items '-alt' in LSF. The reason for the difference between *same* and '-alt' can be understood as arising relatively directly from the lexical semantics of the two operators. In LSF, '-alt' simply requires a plurality of individuals. In English, *same* performs an additional comparison of entities with respect to this plurality of individuals: the additional presupposition associated with *same* is needed to guarantee the existence of such entities.

4.2 Parasitic scope with higher-order traces

In semi-analytic terms, in order to generate the presupposition in (62), *same* needs to be able to access an expression that carries the meaning in (68).

(68) λyx. I would give x answer y

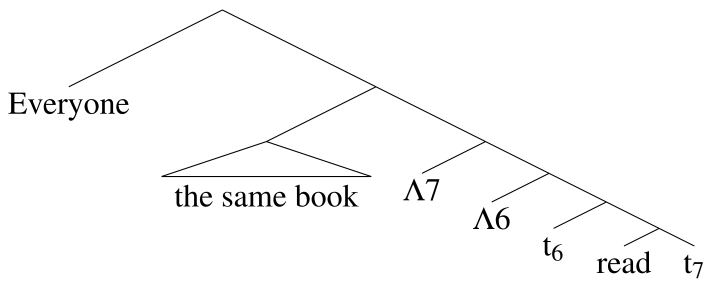
In the event-semantic fragment described in Section 3, however, it turns out that it is impossible to generate this meaning from the event predicate that *same* takes as an argument. Event predicates in fact contain a great deal of information. By manipulating thematic roles, one can, for example, retrieve the set of answers given to any given individual in a particular event. However, the meaning of *would* is not encompassed in any specific giving event; the universal force is constructed by summing multiple giving events. In the event semantic analysis above, *same* thus does not have access to an argument that can provide the meaning in (68).

Stating the problem in this way, parasitic scope appears to much better adapted to an analysis of the relevant presupposition (Barker, 2007). An analysis of parasitic scope is illustrated in (69)-(71). First, the plural licensor of *same* (here, *everyone*) takes scope, thus creating a lambda abstraction (here, Λ6) directly underneath its landing site in the LF. The expression carrying *same* then takes parasitic scope *between* the licensor and its lambda abstraction. In this way *same* has access not just to a predicate (type $\langle e,t\rangle$) as in standard scope-taking, but to a relation (type $\langle e,\langle e,t\rangle\rangle$). *Same* can thus assert that this relation characterizes a constant function: here, the function from people to the books they read is constant.

(69) Everyone read the same book.

(70) $\lambda yx.x$ read book y

(71)

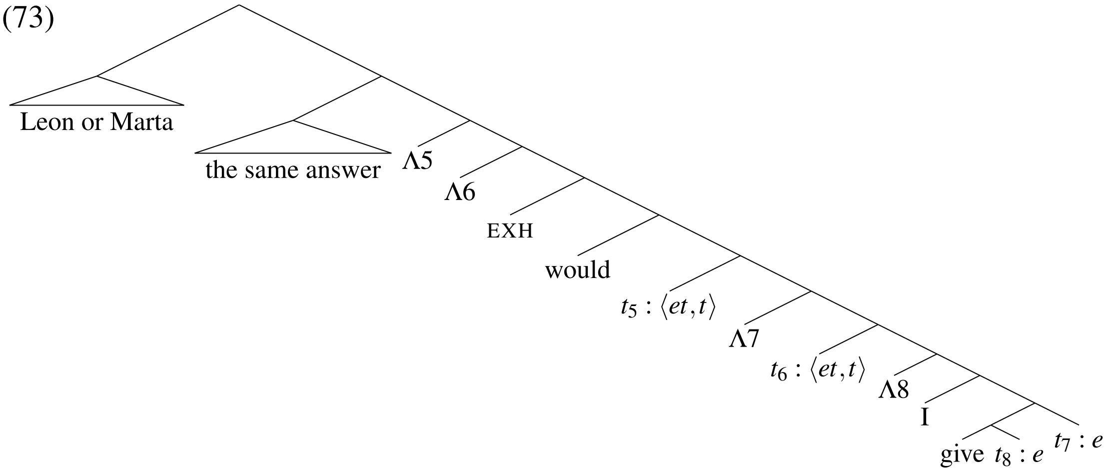


Intuitively, parasitic scope thus gives us exactly the tools we need to access the relation expressed in (68). However, it turns out that parasitic scope also runs into a problem in this example. On the one hand, in order to generate the meaning in (68), *same* must to take parasitic scope on the disjunction, and so the disjunction needs to take scope over *would*. On the other hand, the key observation to the phenomenon of complex licensing is that disjunction is not sufficient by itself to license the internal reading of *same*: it can only do so when it is interpreted in the scope of *would*. Thus, *would* needs to take scope over disjunction. On a standard theory of parasitic scope, these requirements are contradictory: disjunction cannot simultaneously take scope above and below *would*.

A solution to this scope paradox can be provided by allowing disjunction to take split scope by leaving behind a higher order trace (Cresti, 1995; Charlow, 2018; Kuhn, 2022). In this way, disjunction can be interpreted high for the purposes of defining the semantic relation in (68), but can also undergo scope-reconstruction to be interpreted underneath *would*. The tree in (73) thus provides the logical form for (72). The first argument of *the same answer* is thus the expression in (74), denoting a relation between generalized quantifiers.

(72) I would give Leon or Marta the same answer.

(73)



(74) $\lambda F_{\langle et,t \rangle} G_{\langle et,t \rangle}.\text{EXH}(\text{would}'(F(\lambda y.G(\lambda x.\text{give}'(x)(y)(\text{me})))))$

In order for disjunction to serve as a licensor for *same*, its domain of quantification must include a plural individual. We will thus assume that disjunction has an inclusive meaning. Technically, we will say that *Leon or Marta* is a generalized quantifier that quantifies existentially over subparts of the plurality $\text{leon}' + \text{marta}'$, as in (75). Competition with conjunction, implemented here with the exhaustivity operator EXH, will remove the non-atomic subpart from consideration in many contexts, thus preventing disjunction from serving as a licensor by itself.

A few more definitions will be helpful to make our definition of *same* more legible. First, we define a LIFT operator that transforms an individual of type e into a generalized quantifier of type $\langle et,t \rangle$. This operation will be necessary due to the fact that *same* is defined as operating on a relation between generalized quantifiers. Second, we assume that *same* is built out of an indefinite meaning. For legibility, we define *the same answer* in terms of the meaning of the indefinite quantifier *an answer*, in (77).

(75) $[\![\text{Leon or Marta}]\!] = \lambda P.\exists x[x \leq \text{leon}' + \text{marta}' \wedge P(x)]$

(76) $\text{LIFT}(x) = \lambda P.P(x)$

(77) $[\![\text{an answer}]\!] = \lambda P.\exists x[\text{answer}'(x) \wedge P(x)]$

A definition of *the same answer* is provided in (78). The expression takes two arguments; for our target sentence, M corresponds to the expression in (74), and G corresponds to the disjunctive meaning in (75). In (78a), the licensor and the indefinite *an answer* are fed to first argument of *same*, to be interpreted in their reconstructed position. In (78b-d), the definition ensures that the licensor includes a particular predicate. For each individual z in this predicate, (78b) ensures that z is non-atomic, (78c) ensures that, for each atomic part x of z, there is an answer in an M-relation to x, and (78d) ensures that the answers in an M-relation to each atomic part of z are identical. The sentence is false if the condition in (78d) is not met (but the other conditions are), and undefined otherwise.

(78) $[\![\text{the same answer}]\!] = \lambda MG.$

$$
\begin{array}{ll}
1 \text{ if } M([\![\text{an answer}]\!])(G) \wedge & \text{(a)}\\
\quad G(\lambda z.|z| > 1 \wedge & \text{(b)}\\
\quad\quad \forall x \leq_{\text{atom}} z[[\![\text{an answer}]\!](\lambda y.M(\text{LIFT}(y))(\text{LIFT}(x)))] \wedge & \text{(c)}\\
\quad\quad |\{\sigma y[\text{answer}'(y) \wedge M(\text{LIFT}(y))(\text{LIFT}(x))] \;:\; x \leq_{\text{atom}} z\}| = 1) & \text{(d)}\\
0 \text{ if } M([\![\text{an answer}]\!])(G) \wedge & \\
\quad G(\lambda z.|z| > 1 \wedge & \\
\quad\quad \forall x \leq_{\text{atom}} z[[\![\text{an answer}]\!](\lambda y.M(\text{LIFT}(y))(\text{LIFT}(x)))] \wedge & \\
\quad\quad |\{\sigma y[\text{answer}'(y) \wedge M(\text{LIFT}(y))(\text{LIFT}(x))] \;:\; x \leq_{\text{atom}} z\}| \neq 1) & \\
\# \text{ otherwise} &
\end{array}
$$

Let us apply this definition to some examples. First, consider our target sentence in (72). As motivated in Section 3.3, we assume that EXH has no semantic effect with the quantifier *would*. The condition in (78a) thus entails that I would give Leon or Marta an answer—i.e. in all accessible worlds, I give an answer to Leon or Marta or both. Next, the only predicates in the meaning of *Leon or Marta* that satisfy the non-atomicity condition in (78b) are those that include the individual $\text{leon}' + \text{marta}'$; the definition will thus entail that the conditions in (78c) and (78d) apply to this individual. The condition in (78c) entails that for each of Leon and Marta, there is an answer that I would give them. The condition in (78c) entails that the set of answers I would give to Leon is identical to the set of answers I would give to Marta.

In contrast, now consider an equivalent sentence with *have to*, as in (79). Here, we assume that EXH does have an effect, negating the domain alternatives of disjunction. The condition in (78a) thus entails that I have to give Leon or Marta a document, but that I do not have to give Leon a document and that I do not have to give Marta a document. The remainder of the definition runs identically to the sentence with *would*, notably generating the entailment that, for each of Leon and Marta, there is a document that I have to give them. This entailment contradicts the meaning generated from (78a). These contradictory entailments are generated in both the truth and falsity clauses of the definition of *same*, so the resulting sentence is a presupposition failure.

(79) * I have to give Leon or Marta the same document.

A similar logic will apply for simple sentences with disjunction, which we have seen does not license the internal reading of *same* by itself. Differently from our analysis of disjunction in Section 3 (see, e.g., (52)), our analysis here has assumed that disjunction is inclusive, including the sum of the disjuncts in the domain of quantification.[3] The licensing potential of *or* is ruled out by the (obligatory) presence of exhaustification. In (80), for example, the clause in (78a) will entail that Leon or Marta read a book, but not both. This contradicts the clauses in (78b) and (78c), which entail that there is a book that Leon read and a book that Marta read.

(80) * Leon or Marta read the same book.

[3] On the analysis in Section 3, the truth conditions of disjunction are nevertheless inclusive after existential event closure has been applied, but prior to event closure, there are no events in the event predicate that *witness* the inclusive reading.

Finally, we also observe that the definition in (72) is also compatible with more standard cases of the internal reading of *same*. In (81), for example, we can keep a parallel structure with higher-order traces, but in this case, there is no semantic effect of reconstruction, since, unlike (72), there is no semantic material between the general quantifier trace and its binder. The clauses in (78b) and (78c) nevertheless have a non-trivial effect: by entailing an existential predicate of each part of the licensing plurality, the definition rules out a collective reading. This is equivalent to the presupposition of pluractionality observed in Section 3.3, motivated by examples such as (82).

(81) Irene and Angelika read the same book.

(82) * Irene and Angelika wrote the same book.

4.3 Licensing by disjunction under negation

As noted above, the analysis that we have offered in the present section differs from the analysis in Section 3 in that disjunction does include pluralities in its domain of quantification; it is only through competition with conjunction (here, using EXH) that these pluralities are excluded, thus removing the ability of disjunction to license the internal reading of *same*. This fact makes the prediction that licensing by disjunction should become possible in contexts in which exhaustification does not take place—and, in particular, in the context of negation. As seen in (83), this prediction appears to borne out; the sentence carries a meaning on which no pair of students received the same exam—each received a different exam.

(83) I didn't give Leon, Marta or Nicole the same exam.

A slight wrinkle to this observation, though, is that disjunction can only license *same* in DE environments when there are three or more disjunctions: the parallel sentence with two disjuncts in (84a) does not carry an internal reading. Once again, this fact can be explained by competition. Usually, disjunction is logically stronger than conjunction in a downward entailing environment. However, because *same* presupposes a plural argument, the atomic individuals in the domain of disjunction are removed, so disjunction becomes equivalent to conjunction; (84a) is thus blocked by competition with (84b). In such a context, it is only with three or more disjuncts that disjunction becomes logically stronger than conjunction in the scope of negation.

(84) a. * I didn't give Leon or Marta the same exam.
b. I didn't give Leon and Marta the same exam.

4.4 Complex licensing with indefinites beyond disjunction

The story told in Section 3 makes a relatively straightforward prediction: complex licensing should not be specific to disjunction; rather, it should be possible when any indefinite scopes below a distributive quantifier. This prediction, as it turns out, is not the case. Ordinary indefinites like *a student* do not appear to participate in complex licensing. The sentence in (85), for example, is exactly parallel to the earlier sentences with disjunction, but the sentence nevertheless does not allow an internal reading of *same*.

(85) * I would give a student the same answer.

Interestingly, though, not all indefinites act the same way; specifically, noun phrases with *any*, commonly taken to denote indefinites that carry further semantic constraints, do license the internal reading of *same*, as seen in (86). A full analysis must thus capture the variable ability of different indefinites to participate in complex licensing.

(86) I would give any student the same answer.

As it turns out, the analysis provided in the present section provides us with a way to account for the inability for *a*-indefinites to participate in complex licensing, and to model the variation across different indefinites more generally. Specifically, all that needs to be said is that *a*-indefinites are not number neutral in the same was as disjunction; the denotation of *a student*, provided in (87), asserts the existence of an atomic student with a particular property. No predicates in this generalized quantifier consist of exclusively plural individuals, so *a student* is unable to satisfy the plurality condition in (78b). By the same logic, the simplest way to account for the ability of *any*-indefinites to participate in complex licensing is to provide a meaning of *any* that admits non-atomic individuals in the domain of quantification, parallel to the analysis for disjunction, as shown in (88).

(87) $[\![\text{a student}]\!] = \lambda P.\exists x[\text{student}'(x) \wedge P(x)]$

(88) $[\![\text{any student}]\!] = \lambda P.\exists x[{}^{*}\text{student}'(x) \wedge P(x)]$

Support for this proposal comes from the behavior of *a* and *any* under negation. In Section 4.3, we argued that the licensing of *same* by disjunction under negation provided support for the claim that the domain of quantification of disjunction includes plural individuals. For other indefinites, judgments parallel the pattern observed above: *a* cannot license *same* under negation, but *any* can, roughly meaning 'I didn't give any pair of students the same exam'. Again, this can be explained by the assumption that *a student* simply does not have plural individuals in its domain of quantification, but *any student* does.

(89) a. * I didn't give a student the same exam.
b. I didn't any student the same exam.

There are of course many further threads that could be investigated. Notably, the meaning provided in (88) is woefully insufficient to describe the complex behavior of *any* that has been documented in the large literature on negative polarity and free choice. Indeed, one observation that bears mentioning as a thread for future work is the fact that *any* has frequently been described as carrying two components of meaning that, through some mechanism or other, end up being interpreted at different levels in the structure. To give one concrete example, Crnič (2019) argues that *any* has an existential meaning that is interpreted low, and an additive (i.e plural) meaning that is interpreted high. Relatedly, Kuhn (2022) proposes that negative concord items takes split scope using the same compositional mechanisms that are employed here. Combining higher-order *any* and higher-order *same* will lead to interleaved scope, which may offer new possibilities for explaining patterns of licensing.

4.5 Revising the rhetoric

So far, the challenge that patterns of complex licensing pose for syntactic analyses of licensing has been simply stated: when the licensing environment arises from the interaction of two operators, neither of the two can be identified as *the* licensor, carrying the relevant morphosyntactic feature. In the context of the present section, however, this rhetoric must be walked back and the logic of the argument revised. Specifically, on the revised analysis, it is in fact possible to identify a particular syntactic element as the licensor: it is the element on which *same* takes parasitic scope—here, this is disjunction.

In light of this revised analysis, the challenge for a syntactic theory of licensing is then the following: if disjunction sometimes bears a morphosyntactic feature that allows it to license distributive concord items, why is this feature not always available? A possible answer to this question can be built on the analysis of Ivlieva (2013), who argues based on number agreement that disjunctions can bear either singular or plural number, but that the distribution of plural number marking itself has a distribution determined by a semantic constraint arising from obligatory exhaustification. An agreement theory of distributive concord could thus in principle posit that the plural marker -PL bears a feature [iDist] that can syntactically license distributive concord items, but that -PL is itself subject to semantically-determined distributional constraints when it appears on disjunction.

It is possible that such a theory can be made to work. On the other hand, the syntactic theory becomes significantly less parsimonious when it is forced to posit that syntactic licensors must themselves be semantically licensed. Moreover, as sketched, this revised syntactic analysis has no explanation for the inability for *have to* or temporal quantifiers to participate in complex licensing of *same*, as they should in principle create the semantic environment needed for the presence of -PL. In contrast, the semantic presuppositions of *same* on the theory of semantic licensing are rather well grounded in the lexical semantics of *same*: in order to make a comparison of sameness, one needs to presuppose the existence of multiple things that can be compared.

4.6 Comparing event semantics and parasitic scope

In the present paper, we have presented two analyses of complex licensing that use fundamentally different frameworks. In Sections 2-3, an analysis was presented within event semantics; Section 4 has presented an analysis in terms of parasitic scope. This shift in implementation was motivated by empirical considerations, but also permits us to explore the relation between the two frameworks.

Both frameworks have antecedents in the literature. Event semantics has rather naturally been used in the literature on pluractionals (Lasersohn, 1995), and was notably also proposed by Carlson (1987) for the analysis of English *same*. Parasitic scope was famously proposed to give the first compositional semantic analysis of *same* (Barker, 2007). Interestingly, Carlson (1987) himself does not provide a compositional analysis of *same* in event semantics ("its formal implementation requires extensive work"), and Barker (2007)—while explicitly building on many insights from Carlson (1987)—ultimately provides an analysis without events. The present paper thus provides what is perhaps the first compositional implementation of the event-semantic proposal of Carlson (1987). Formally, two key elements that permitted this implementation were (a) a mechanism for an item (here, *same*) to take split scope and (b) a mechanism that allows individuals to be associated across the two scopal levels. In event semantics, the first of these ideas is foreshadowed in the scopal analysis of pluractionality by Kuhn and Aristodemo (2017), but the most complete antecedents of the implementation here come from yet another framework: a body of literature in dynamic semantics to analyze dependencies involved in sentence-internal phenomena (Henderson, 2014; Charlow, 2018; Bumford, 2017; Kuhn, 2022). As discussed in Section 3.4, however, these analyses in many cases do not fundamentally hinge on the use of dynamic representations, so can be translated into a static framework that has similar representational properties, such as event semantics, in which thematic roles play the role of the indices in dynamic semantics.

Having two complete implementations in separate frameworks also allows us to compare the two. Notably, any analysis of *same* relies on the fundamental insight that *same* operates on a two-place relation (and not a one-place predicate, like generalized quantifiers). The mechanism of parasitic scope furnishes this relation directly, using a structural means to provide *same* with an expression with two lambda abstractions. Within event semantics, for most cases, the relevant relation can be defined with respect to thematic roles of the parts of an event plurality: *same* checks that the set of values taken by thematic role A is constant when considering each maximal subpart of the event that has a particular value for thematic role B.

Of course, as we saw in Section 4.2, defining this relation with respect to thematic roles is in fact insufficient in the case of complex licensing. The reason for this is that the meaning of a distributive operator is not encapsulated in any single event, but is witnessed by the existence of a sum of events. It is notable, though, that the challenge facing the event-based analysis only appears in very precise circumstances: it only occurs for *same*, which has more presuppositions than pluractional markers, and it only occurs in cases of complex licensing, which depend on the presence of a distributive operator which is not itself the licensor of *same*.

The analyses of complex licensing using event semantics and parasitic scope also turn out to share an important formal property. Empirically, complex licensing is distinguished by the fact that the plurality of entities that licenses the distributive concord item is introduced by an item that is necessarily interpreted below another distributive operator. On both analyses, one thus needs a mechanism of split scope: allowing a single item to introduce a meaning below the distributive operator, but also make a semantic contribution above it. This is the reason why the parasitic scope analysis, which is already scopal in nature, had to be further enriched with split scope, to allow the disjunctive meaning to be reconstructed below the modal quantifier. This then appears to be a formal property of complex licensing that transcends frameworks: on any analysis, the phenomenon necessitates some way of taking split scope.

5 Conclusion

In this paper, we have introduced the new phenomenon of *complex licensing*, in which distributive concord items are licensed by the interaction of multiple operators in a sentence, none of which can serve as a licensor alone. We have illustrated the pattern for two different phenomena in two different languages: participant-key pluractionals in LSF and the internal reading of *same* in English. In both cases, the items can be licensed by the interaction of a distributive operator, disjunction, and, depending on the analysis, an exhaustivity operator.

We have provided fully compositional, semantic analyses of the phenomena in question. The analytical implementations were carried out two frameworks. First, we provided an analysis using event semantics, including what is perhaps the first compositional implementation of Carlson's (1987) analysis of *same*. Second, in order to capture a wider set of data, we provided an analysis of the complex licensing of *same* using parasitic scope (Barker, 2007).

These phenomena have allowed us to tease apart various theoretical approaches to licensing. Most notably, the phenomenon of complex licensing poses great challenges to syntactic theories of licensing, which require a single element in the sentence to serve as *the* licensor, bearing a relevant morphosyntactic feature. In a best-case scenario for a syntactic theory, it may be possible to identify one of the multiple interacting elements as the syntactic licensor, but the presence of the relevant morphosyntactic feature must itself be subject to semantic licensing.

Within semantic theories, we have discussed the relation between event semantics and dynamic semantics, as well as the relation between an analysis in terms of event semantics and an analysis in terms of parasitic scope. We observed that the particular event-semantic analysis presented here is unable to capture the full subtleties of the licensing of English *same*, due to the way that distributivity is standardly modeled in event semantics. On the other hand, for much empirical landscape, the two analyses converge, and indeed the phenomenon of complex licensing appears to necessitate certain formal properties that transcend frameworks; in particular, on both analyses, there is a central role for split scope.

Conflicts of interest

The author(s) declare no conflicts of interest.

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